



GBA-Related Biomarkers in Parkinson's disease Analysis of β -Glucocerebrosidase Activity

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Summary (or Abstract)

Mutations in the β -glucocerebrosidase gene (GBA) modulate lysosomal accumulation of β -glucocerebrosidase and Parkinson's disease progression. Lower activity levels of β -glucocerebrosidase were reported in dried blood spots prepared from fresh blood of Parkinson's disease (PD) patients [1]. We have quantified β -glucocerebrosidase activity by tandem mass spectrometry (MS/MS) in dried blood spots prepared from previously frozen whole blood.

Method

Spotting of previously frozen whole blood

Whole blood obtained from PPMI was thawed on watered ice gently rocking for 45- 60 mins. Once thawed, 75uL of blood was spotted on Whatman 903™ filter paper (GE Healthcare, Piscataway, NJ) dropwise then dried at room temperature overnight. Samples were dried in horizontal position in a well ventilated area. Dried blood spots were sealed in plastic bags with desiccant (Uline, Pleasant Prairie, WI) and humidity indicator cards (Static Control, Sanford, NC) and stored at -80°C until analysis.

β -glucocerebrosidase Activity Analysis

Samples were analyzed as previously described for dried blood spots prepared from fresh blood [2-4]. In brief, blood extract from one 3.2 mm punch per DBS sample was combined with C12-glucocerebroside substrate and C14-ceramide internal standard mixtures. Sealed 96-well plates were incubated on an orbital shaker at 37°C for 20-24 hours. Quenched samples were cleaned up by liquid-liquid and solid phase extraction. Dried sample plates were stored at -80°C. Prior to MS/MS analysis, plates were thawed and reconstituted with 200µl of mobile phase. Analytes were monitored on an API5000 triple quadrupole mass spectrometer in positive mode by MRM. The enzyme activity of each sample was calculated from the ion abundance ratio of product to internal standard as measured by the mass spectrometer. Background activity of blank filter paper was subtracted from the DBS activity. Activity was expressed as micromoles of product per liter of whole blood per hour (µM/h). QC samples with previously established activity levels for β -glucocerebrosidase were included in each plate as assay controls.





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